

EPILEPSY DIGITAL TWIN REPORT • INTEGRATED NEUROLOGICAL DECISION SUPPORT

Neuro Digital Twin Integrated Clinical Report

An integrated clinical assessment combining follow-up visits, seizure history, EEG, ECG, CT/MRI imaging, laboratory results, and pharmacogenomic testing for Focal impaired-awareness seizures.

Patient & Clinical Context		9/9 data categories included · generated 2026-08-28T15:43:28				
PATIENT John Doe Patient ID SYN-100482	AGE / SEX 34 years · Male	PRIMARY DIAGNOSIS Focal impaired-awareness seizures Secondary diagnosis: Secondary generalized tonic-clonic seizures	DIAGNOSED SINCE 12 May 2021			
CURRENT CLINICAL STATUS Stable outpatient epilepsy state with persistent breakthrough seizure risk						
CURRENT THERAPY Levetiracetam (Keppra) 1000 mg BID · Lamotrigine 150 mg BID						
REPORTING PERIOD Data reviewed: 01 Jan – 30 Jun 2026 (~6 months) Clinic visit notes cover 01 Jan 2026 to 02 Mar 2026; laboratory, imaging, EEG/ECG, and questionnaire data continue through 30 Jun 2026.						
5 Clinical Visits	12 CBC Reports	6 CT Studies	6 MRI Series	12 ECG Sessions	35 EEG Recordings	12 Questionnaires

01 Executive Summary

Across five clinical-note visits from 2026-01-01 to 2026-03-02, the patient has chronic focal epilepsy with persistent breakthrough seizures on levetiracetam and lamotrigine. The clinician repeatedly documented reasonably stable seizure control and favorable quality-of-life trajectory, while seizure burden, sleep quality, trigger exposure, aura symptoms, and medication tolerability continued to fluctuate.

67%
Overall Health Score

Moderate
Overall Clinical Risk

Elevated
EEG-Derived Seizure Risk

10 (7 high)
Genetic Safety Flags

Domain the patient rates worst on questionnaires: Epileptic Triggers. See the Digital Twin Risk Dashboard (Section 13) for the full risk breakdown.

02 6-Month Timeline

01 JAN 2026

Routine epilepsy follow-up with stable clinician impression — Contributes to longitudinal assessment of seizure burden, triggers, sleep, and tolerability.

16 JAN 2026

Seizure clustering documented during follow-up — Raises near-term seizure-monitoring priority despite otherwise stable clinician impression.

03 FEB 2026

CBC/LFT safety review documented — Supports medication-safety monitoring without a documented abnormal lab signal.

15 FEB 2026

Highest recorded seizure burden in clinical-note window — 11 reported episodes make this the peak burden visit.

02 MAR 2026

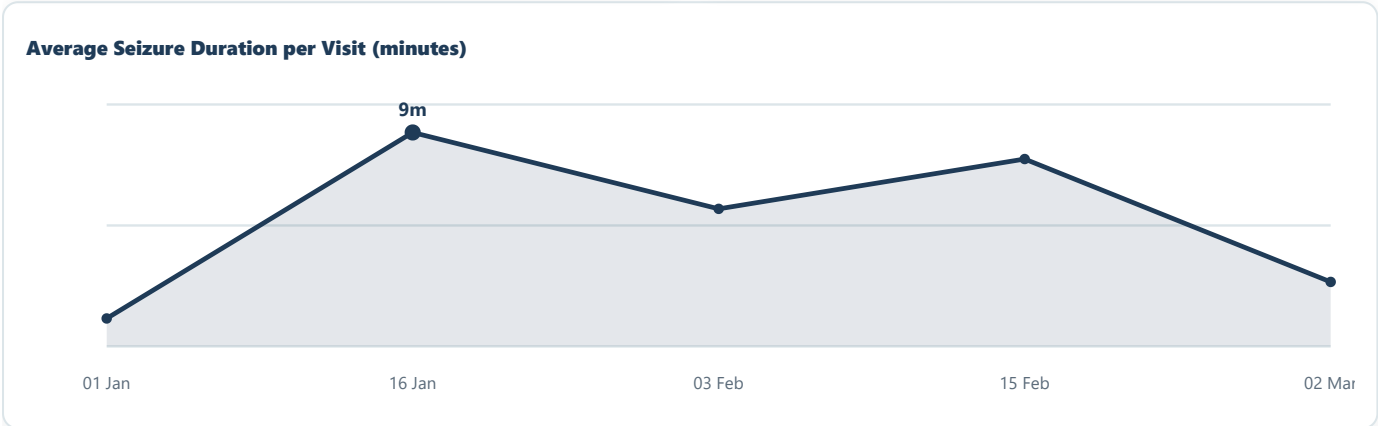
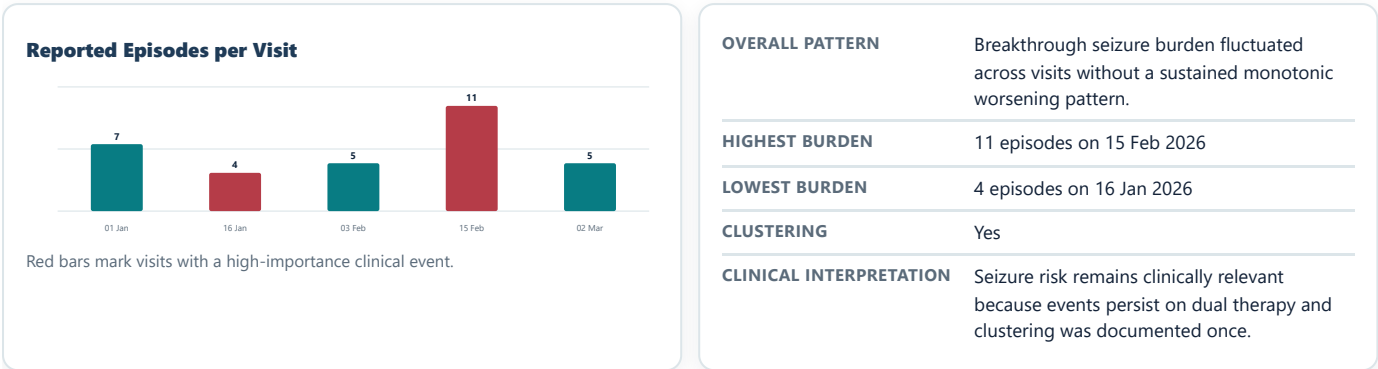
Routine epilepsy follow-up with stable clinician impression — Contributes to longitudinal assessment of seizure burden, triggers, sleep, and tolerability.

28 AUG 2026

Pharmacogenomic (PGx) panel reported — 76 variants analyzed across 47 medications; see Pharmacogenomics.

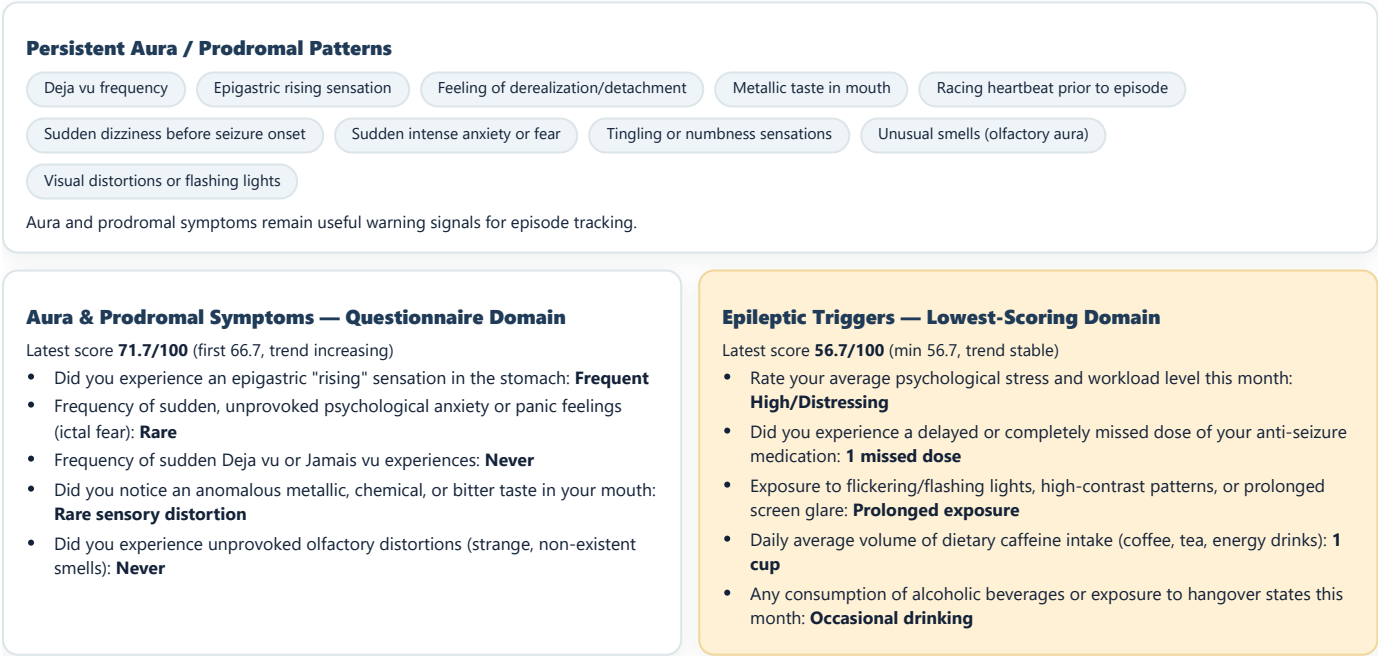
03 Seizure Trends

Clinical-note episode counts, clustering, and duration



04 Aura & Triggers

Prodromal patterns and patient-reported trigger exposure



05 Medication Response

Current regimen, tolerability, and reported side effects

Current Regimen

DRUG	DOSE
Levetiracetam (Keppra)	1000 mg BID
Lamotrigine	150 mg BID

EFFECTIVENESS

Partial response

TOLERABILITY

Generally tolerated with intermittent dizziness and variable side-effect burden

DOCUMENTED TOXICITY

No

CLINICAL INTERPRETATION

Current antiseizure therapy is associated with stable clinician impressions but incomplete seizure suppression.

Persistent Side Effects (clinical notes)

Memory fog post-dose

Tremor in hands

Weight change since last visit

Gum hypertrophy

Mood irritability

Nausea after taking medication

Skin rash or itching

Daytime drowsiness

Dizziness/lightheadedness

Unsteady gait (ataxia)

Medication Side Effects — Questionnaire Domain

Latest score **78.3/100** (first 68.3, trend increasing)

• Rate the severity of daytime drowsiness or somnolence: **Mild (Manageable)**

• Experience of dizziness, vertigo, or lightheadedness after taking medication: **Never**

• Presence of physical instability, unsteadiness, or clumsy movements (ataxia): **Slight imbalance**

• Frequency of unprovoked irritability, anger, or mood volatility: **Mild irritability**

• Signs of dermatological distress or unexplained skin rashes: **Clear skin**

06 Pharmacogenomics

76 variants across 47 medications · reported 28 Aug 2026

PGx summary for the current regimen: the single most clinically significant finding per drug, with its genetic basis and source citation.

DRUG	GENE (GENOTYPE)	WORST / MOST SIGNIFICANT OUTCOME	CITATION
Lamotrigine	CYP3A4 (TT)	REDUCED EFFICACY	PMID:28343093
Levetiracetam	SCN1A (CT)	EFFICACY	PMID:28753467

07 Drug Interactions

2 current medication(s) reconciled · Flockhart Cytochrome P450 Drug Interaction Table (curated-snapshot-2024)

Medication Reconciliation

DRUG	DOSE	STATUS	FLAGS
Lamotrigine	150.0 mg BID	current	—
Levetiracetam (Keppra)	1000.0 mg BID	current	—

No conflicting doses found for the same drug across the reconciled source data.

Clinically Relevant Interactions

No clinically required drug-drug interaction was identified for the current regimen.

Essential Monitoring

Continue EEG-derived seizure-risk monitoring.

Continue periodic CBC and safety-laboratory monitoring.

Continue seizure frequency, duration, and clustering tracking.

Review the priority safety flags in the Pharmacogenomics section before adding or changing an interacting medication.

Limitations of this screen: This assessment is clinical decision support, not an automated prescribing system; no dose, start, or stop decision is made or implied here. No proposed/candidate medication list is present in the source data, so only the current regimen's pairwise and therapy-level evidence is evaluated. Curated pharmacokinetic (CYP) coverage is limited to a small bundled reference table; a drug not listed there is reported as unresolved, not as free of interactions.

Generated from validated patient data. Decision-support summary only -- not a substitute for clinician review.

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35
Total Recordings

12
Interictal

11
Preictal

12
Ictal

Signal Variance Midpoint by Phase (µV²)

Phase	Signal Variance Midpoint (µV²)
Interictal	344.45
Preictal	4343.59
Ictal	42395.2

EEG-derived risk: Elevated

Future seizure probability: Elevated

Dominant feature: Signal Variance

Phase Analysis

PHASE	STATE	ACTIVITY	VARIABILITY
Interictal	Baseline	Stable	Low
Preictal	Transition	Increasing Instability	Moderate-High
Ictal	Seizure	Hypersynchronous	Very High

PHASE	AMPLITUDE RANGE	STD DEV RANGE	VARIANCE RANGE
Interictal	-89; 99	11.68; 23.5	136.44; 552.46
Preictal	-345; 440	14.48; 92.07	209.71; 8477.48
Ictal	-504; 784	52.5; 286.42	2756.47; 82033.96

Longitudinal EEG recordings demonstrate a consistent transition from stable interictal activity to preictal electrical instability followed by high-amplitude ictal seizure activity. Findings are consistent with the dataset's interictal/preictal/ictal phase taxonomy, indicating reproducible seizure-related signal evolution; this is a signal-classification observation, not a clinician-verified diagnosis.

CT — 6 studies, 12 Jan – 29 Jun 2026

DATE	IMPRESSION
12 Jan 2026	No acute intracranial abnormality is evident on this single-image model-assisted review.
16 Feb 2026	No gross acute abnormality is evident on this limited skull-base level image; assessment is limited by slice level and image quality.
11 Mar 2026	No acute intracranial abnormality is evident on this single-image model-assisted review.
20 Apr 2026	No acute intracranial abnormality is evident on this high-convexity image; ventricular assessment is not available at this slice level.
14 May 2026	No acute intracranial abnormality is evident on this limited-quality near-vertex image.
29 Jun 2026	No acute intracranial abnormality is evident. Incidental bilateral choroid plexus calcifications are present.

Across the available CT image files, no gross acute intracranial hemorrhage, midline shift, or mass effect is evident on this model-assisted image-only review. Direct progression assessment is limited because the files represent single slices with different slice levels, resolutions, and image quality. This output is not a radiologist-verified diagnostic report.

MRI — Metadata-supported structural stability, diagnostic interpretation not available

FINDING	STATUS	DETAIL
Brain Morphology	NOT DIRECTLY ASSESSED	The provided files include volumetric MRI metadata, but no radiologist interpretation or morphometric segmentation was available for direct morphology assessment.
Brain Volume	STABLE WITHIN AVAILABLE METADATA	T1w nonzero volume range: 8387642 to 8777268 mm3 (mean 8641065 mm3; range 4.5%). FLAIR nonzero volume range: 6022385 to 6625690 mm3 (mean 6231344 mm3; range 9.7%). These are preprocessing-derived nonzero voxel volumes, not formal brain volumetry.
Tissue Signal Characteristics	QUANTITATIVELY SUMMARIZED, NOT DIAGNOSTICALLY INTERPRETED	FLAIR mean intensity 28.76 to 32.27; standard deviation 47.57 to 54.16; T1w mean intensity 99.82 to 105.56; standard deviation 153.13 to 162.05. Intensity statistics are useful for preprocessing consistency but are not diagnostic tissue-signal interpretation.

Not directly assessed from the available imaging metadata (this means "not evaluated," not "abnormal"): White Matter, Hippocampus, Cortical Structure, Mass Lesion, Hemorrhage, Infarction.

Stable by available preprocessing metadata. 6 MRI series were available across 3 study identifiers, including 3D T1-weighted MRI, T2-FLAIR MRI. Preprocessed metadata show consistent 3D acquisition characteristics and no metadata-level evidence of major global structural progression. A radiology report, lesion segmentation, hippocampal volumetry, and cortical thickness analysis were not available, so subtle pathology cannot be excluded from these metadata alone.

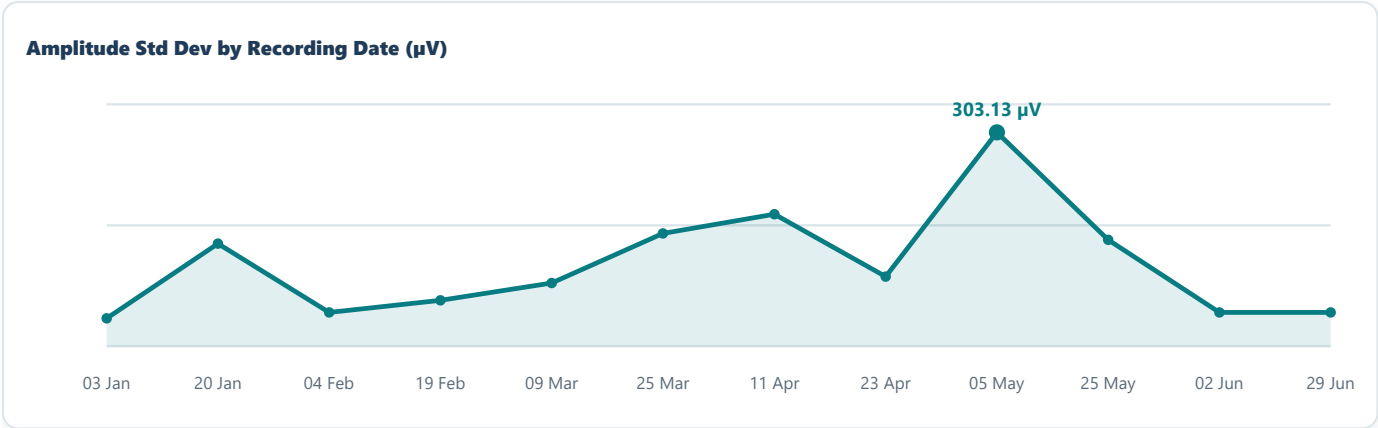
10ECG

single channel ("ECG SD") in every recording

12
Recordings

99h 11m
Total Duration

256
Sampling Rate (Hz)



- All 12 recordings use a single channel sampled at 256 Hz, so statistics are directly comparable across sessions.
 - Recording duration varies considerably, from about 1h 11m (2026-04-11) to about 22h 27m (2026-03-25).
 - The 2026-06-02 and 2026-06-29 recordings have matching percentile values and near-identical histogram shapes, suggesting very similar signal characteristics or a data duplication between the two sessions.
 - Standard deviation of amplitude ranges from about 82.6 uV (2026-01-03) to about 303.1 uV (2026-05-05), indicating substantially different signal variability/noise levels between sessions.
 - Peak-to-peak amplitude (max minus min) ranges from roughly 2,483 uV (2026-05-25) to about 3,577 uV (2026-03-25).
- This report is derived strictly from the statistical fields present in the source dataset. It does not represent a clinical or diagnostic ECG interpretation. Any assessment of cardiac rhythm, heart rate, or abnormality requires review of the underlying waveform by a qualified clinician or an appropriately validated diagnostic algorithm, neither of which is reflected in this report.

11CBC / Labs

12 reports, 01 Jan – 30 Jun 2026

Overall hematological status: Stable within reference ranges. No CBC parameter abnormality was identified against the report-provided reference ranges across the observation period.

PARAMETER	FIRST	LATEST	RANGE	TREND
Hemoglobin (Hb)	14.4	15.1	13.5 – 15.9	increasing
Total RBC count	5.0	5.1	4.8 – 5.3	increasing
Packed Cell Volume (PCV)	44.7	43.5	42.2 – 48.4	decreasing
Mean Corpuscular Volume (MCV)	90.0	97.8	88.4 – 98.8	increasing
MCH	27.6	29.9	27.6 – 31.1	increasing
MCHC	33.1	33.0	32.9 – 34.3	decreasing
RDW	12.1	13.5	12.1 – 13.5	increasing
Total WBC count	5761.0	9630.0	5476.0 – 9973.0	increasing
Neutrophils	58.0	54.0	52.0 – 61.0	decreasing
Lymphocytes	22.0	29.0	22.0 – 34.0	increasing
Eosinophils	4.0	4.0	1.0 – 5.0	stable
Monocytes	3.0	3.0	1.0 – 8.0	stable
Basophils	1.0	1.0	0.0 – 2.0	stable
Platelet Count	251725.0	209427.0	189551.0 – 373102.0	decreasing

Patient-reported vs. measured discrepancy: the 30 Jun 2026 questionnaire records patient recollections of White Blood Cell (WBC) count baseline trend from your latest Complete Blood Count (CBC): Mild Leukopenia (Borderline low); Platelet Count profile stability from recent blood evaluations: Borderline Thrombocytopenia (Low); Liver Function Test (LFT) - SGOT/AST status: Mild Elevation (40-100 U/L); measured CBC values across all 12 reports in this period remained within their reference ranges throughout (see table below). The discrepancy likely reflects patient recall/perception rather than a measured lab abnormality.

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Sleep, Cognition & Quality of Life

Sleep		Cognition		Quality of Life	
TREND	Sleep remained variable across visits	MEMORY	Stable on clinician review, with low to intermittent patient-reported memory difficulty	DAILY FUNCTION	Independence in daily activities was repeatedly normal, intact, or stable
AVERAGE	6.2 hours per night across printed assessments	ATTENTION	Variable patient-reported concentration and attention symptoms	SOCIAL FUNCTION	Intermittent social withdrawal or isolation symptoms were reported
Daytime naps needed		LANGUAGE	Mild word-finding difficulty was noted but not described as worsening	CLINICAL INTERPRETATION	Quality of life remains affected by seizures, sleep, triggers, and medication tolerability, despite generally stable outpatient status.
Difficulty falling back asleep		MENTAL CLARITY	Improved mental clarity was reported at the latest visit		
Fatigue level through the day					
Waking up unrefreshed					

Latest Questionnaire Domain Scores (0-100)		
Sleep Quality & Architecture		91.7/100
Cognitive & Executive Function		83.3/100
Lab Biomarkers & Lifestyle		75/100

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Digital Twin Risk Dashboard

Overall clinical risk: Moderate

Future Seizure Risk Moderate	Status Epilepticus Risk Low To Moderate	Hospitalization Risk Low To Moderate	Cognitive Decline Risk Low
Medication Toxicity Risk Low To Moderate	Functional Decline Risk Low	EEG-Derived Seizure Risk Elevated	Genetic Safety Flags 10 Flags, 7 High

State Vector (Health Metric Scores, 0-100)		
Disease Stability		67/100
Treatment Response		64/100
Medication Safety		67/100
Cognitive Health		69/100
Sleep Health		66/100
Quality Of Life		67/100
Each domain score is the mean of clinician-documented per-visit ratings across the clinical-note window; overall health score is the mean of the other domains. Derived digital-twin summary metrics, not a direct lab/imaging measurement.		

14 Clinical Action Plan

Priority Actions (Pharmacogenomics)

- HIGH** **Avoid or use with heightened monitoring**
See priority_safety_flags for the specific gene-drug pairs (ABCB1 (CC), BDNF (TC), CREB1 (AA), FKBP5 (TC), GRIA3 (A), HTR1B (TA), METTL21A (AA)).
- MODERATE** **Dose with caution / consider therapeutic drug monitoring**
See metabolizer_profile for reduced-function genes (CYP2C9, CYP2B6, CYP3A5, UGT1A4, POR).
- INFORMATIONAL** **Contextualize**
Interpret alongside actual clinical history, current medication list, and labs -- this panel alone does not establish a diagnosis.

Recommended Actions

- continue the documented antiseizure medication regimen unless changed by the treating clinician
- maintain the documented plan for consistent sleep schedule and trigger avoidance
- continue seizure and aura tracking across follow-up windows
- continue periodic CBC, liver function, and antiseizure drug-level safety monitoring when scheduled
- reassess at the planned follow-up interval

Monitoring Priorities

- seizure frequency by type
- seizure clustering
- average and longest seizure duration
- post-ictal recovery time
- aura pattern and severity
- sleep duration and sleep fragmentation
- missed medication doses and trigger exposure
- dizziness and other medication side effects
- CBC, liver function, and antiseizure drug-level tracking
- word-finding, memory, and attention symptoms
- Signal Variance
- Amplitude Dynamics
- Electrical Stability
- Preictal Detection
- Seizure Frequency
- Temporal Disease Progression

MRI Follow-Up

- Perform formal radiology review of the MRI image volumes.
- Run automated brain segmentation and longitudinal registration across the three study identifiers.
- Perform hippocampal volumetry because epilepsy-related mesial temporal abnormalities cannot be excluded from metadata alone.
- Run FLAIR white matter lesion analysis if clinically relevant.
- Compare future MRI studies using the same T1w and FLAIR acquisition strategy when possible.
- Correlate MRI findings with seizure semiology and neurological examination.
- Review MRI together with EEG findings for seizure-focus localization.
- Repeat or escalate imaging review if new neurological deficits, worsening seizures, or new focal symptoms occur.

Medication Reconciliation

- Current regimen reconciled and checked against curated reference pharmacology and this patient's own pharmacogenomic findings (Section 07). Predicted-model (SuperCYPsPred) coverage is still unresolved, so a live, comprehensive DDI screen is recommended before adding or changing any interacting medication.